

Overview of Logistic regression

Subject: Logistic regression

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$P(y = 1 | x) = \sigma(w^T x + b)$$

1.

$$g(p(x)) = \sigma^{-1}(p(x)) = \text{logit} \quad p(x) = \ln \left(\frac{p(x)}{1-p(x)} \right) = \beta_0 + \beta_1 x, \quad \{\displaystyle g(p(x)) = \sigma^{-1}(p(x)) = \operatorname{logit} \quad p(x) = \ln \left(\frac{p(x)}{1-p(x)} \right) = \beta_0 + \beta_1 x, \}$$

2.

Definition of the inverse of the logistic function We can now define the logit (log odds) function as the inverse $g = \sigma^{-1}$ of the standard logistic function. It is easy to see that it satisfies:

3.

$\beta_0 = -3$ is the y-intercept. It is the log-odds of the event that $y = 1$, when the predictors $x_1 = x_2 = 0$. By exponentiating, we can see that when $x_1 = x_2 = 0$ the odds of the event that $y = 1$ are 1-to-1000, or 10^{-3} . Similarly, the probability of the event that $y = 1$ when $x_1 = x_2 = 0$ can be computed as $1 / (1000 + 1) = 1 / 1001$.

4.

$$\begin{aligned} \varepsilon_0 &\sim \text{EV}_1 \blacksquare (0, 1) & \varepsilon_1 &\sim \text{EV}_1 \blacksquare (0, 1) \\ \operatorname{EV}_{-1}(0,1) &\backslash\!\!\backslash \varepsilon_1 & \operatorname{EV}_{-1}(0,1) &\backslash\!\!\backslash \varepsilon_1 \end{aligned}$$

5.

$$p_n(x) = \frac{e^{\beta_n \cdot x_1 + \sum_{u=1}^N \beta_u \cdot x_u}}{e^{\beta_n \cdot x_1 + \sum_{u=1}^N \beta_u \cdot x_u} + 1} \quad \text{for } n = 1, 2, \dots, N$$

6.

where $\Delta(n, y_k)$ is an indicator function which equals 1 if $y_k = n$ and zero otherwise. In the case of two explanatory variables, this indicator function was defined as y_k when

$n = 1$ and $1 - y_k$ when $n = 0$. This was convenient, but not necessary. Again, the optimum beta coefficients may be found by maximizing the log-likelihood function generally using numerical methods. A possible method of solution is to set the derivatives of the log-likelihood with respect to each beta coefficient equal to zero and solve for the beta coefficients:

7.

This functional form is commonly called a single-layer perceptron or single-layer artificial neural network. A single-layer neural network computes a continuous output instead of a step function. The derivative of p_i with respect to $X = (x_1, \dots, x_k)$ is computed from the general form:

8.

If the model deviance is significantly smaller than the null deviance then one can conclude that the predictor or set of predictors significantly improve the model's fit. This is analogous to the F-test used in linear regression analysis to assess the significance of prediction.

9.

$$p_0(x) = 1 - \sum_{n=1}^N p_n(x) = \frac{1}{1 + \sum_{u=1}^N e^{\beta_u \cdot x}} \quad \{ \displaystyle p_0(x) = \frac{1}{1 + \sum_{u=1}^N e^{\beta_u \cdot x}} \}$$

10.

General Logistic regression is used in various fields, including machine learning, most medical fields, and social sciences. For example, the Trauma and Injury Severity Score (TRISS), which is widely used to predict mortality in injured patients, was originally developed by Boyd et al. using logistic regression. Many other medical scales used to assess severity of a patient have been developed using logistic regression. Logistic regression may be used to predict the risk of developing a given disease (e.g. diabetes; coronary heart disease), based on observed characteristics of the patient (age, sex, body mass index, results of various blood tests, etc.). Another example might be to predict whether a Nepalese voter will vote Nepali Congress or Communist Party of Nepal or for any other party, based on age, income, sex, race, state of residence, votes in previous elections, etc. The technique can also be used in engineering, especially for predicting the probability of failure of a given process, system or product. It is also used in marketing applications such as prediction of a customer's propensity to purchase a product or halt a subscription, etc. In economics, it can be used to predict the likelihood of a person ending up in the labor force, and a business application would be to predict the likelihood of a homeowner defaulting on a mortgage. Conditional random fields, an extension of logistic regression to sequential data, are used in natural language processing. Disaster planners and engineers rely on these models to predict decisions taken by householders or building occupants in small-scale and large-scales evacuations, such as building fires, wildfires, hurricanes among others. These models help in the development of reliable disaster managing plans and safer design for the built environment.

11.

$$\Pr(Y_i = c) = \frac{e^{\beta_c \cdot X_i}}{\sum_{h=1}^H e^{\beta_h \cdot X_i}} \quad \{ \displaystyle \Pr(Y_i = c) = \frac{e^{\beta_c \cdot X_i}}{\sum_{h=1}^H e^{\beta_h \cdot X_i}} \}$$

12.

$$\begin{aligned} \Pr(Y_i = 0) &= e^{\beta_0 \cdot X_i} / (e^{\beta_0 \cdot X_i} + e^{\beta_1 \cdot X_i}) \\ \Pr(Y_i = 1) &= e^{\beta_1 \cdot X_i} / (e^{\beta_0 \cdot X_i} + e^{\beta_1 \cdot X_i}) \end{aligned}$$

13.

The regression coefficients $\beta_0, \beta_1, \dots, \beta_m$ are grouped into a single vector β of size $m + 1$. For each data point i , an additional explanatory pseudo-variable $x_{0,i}$ is added, with a fixed value of 1, corresponding to the intercept coefficient β_0 . The resulting explanatory variables $x_{0,i}, x_{1,i}, \dots, x_{m,i}$ are then grouped into a single vector X_i of size $m + 1$. This makes it possible to write the linear predictor function as follows: